Name: ____________________

WORKSHEET

Expanding algebraic expressions

Ever since I was a child, I have had this instinctive urge for expansion and growth. To me, the function and duty of a quality human being is …

14	8	15

7	11	13	9	15	10	15

1	13	2

8	12	13	15	7	14

2	15	4	15	6	12	5	16	15	13	14

12	3

12	13	15	7

5	12	14	15	13	14	11	1	6

Bruce Lee

Rules

$a(b + c) = ab + ac$

$(a + b)(c + d) = ac + ad + bc + bd$

$(a + b)(a - b) = a^2 - b^2$

$(a + b)^2 = a^2 + 2ab + b^2$

$(a - b)^2 = a^2 - 2ab + b^2$

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Questions

Expand the following algebraic expressions.

1 $3(x + 2) =$

2 $2y(3y - 7) =$

3 $-5x(7x + 3y - 2) =$

4 $(x + 3)(x - 2) =$

5 $(3x - 1)(2x + 5) =$

6 $(x + y)(x - y) =$

7 $(x - 7)(x + 7) =$

8 $(x + 5)^2 =$

9 $(2y - 3)^2 =$

10 $(x + y)^2 =$

11 $(3x + 4)(2 - 5x) =$

12 $(7x + 2)(3x - 2) =$

13 $(4x + 1)^2 =$

14 $(2y - 3)(2y + 3) =$

15 $(x + 8)(x - 8)(x + 1) =$

16 $(4x + 2)(3x - 1)(2x + 5) =$

Solutions

A $3x + 6$

C $4y^2 - 12y + 9$

D $6y^2 - 14y$

E $x^3 + x^2 - 64x - 64$

F $-35x^2 - 15xy + 10x$

H $x^2 + 10x + 25$

I $-15x^2 - 14x + 8$

L $x^2 - y^2$

M $24x^3 + 64x^2 + 6x - 10$

N $16x^2 + 8x + 1$

O $21x^2 - 8x - 4$

P $6x^2 + 13x - 5$

R $x^2 + 2xy + y^2$

S $x^2 - 49$

T $4y^2 - 9$

V $x^2 + x - 6$

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